[INCH-POUND]
T-M-561G
February 11, 1991
SUPERSEDES
T-M-561F
September 20, 1977

FEDERAL SPECIFICATION

MOPHEAD, WET
(COTTON-SYNTHETIC AND SYNTHETIC)

This Federal Specification has been approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers wet use floor cleaning mopheads with taped and untaped cut ends, and taped looped ends, composed of cotton-synthetic and synthetic yarns.

1.2 Classification.

1.2.1 Type I. Type I mopheads shall be cut end mopheads of the following styles as specified (see 6.2).

1.2.1.1 Styles.
Style 1 - Untaped, 8 +/- 2 ply yarn.
Style 2 - Taped, 4 ply minimum yarn.
Style 3 - Untaped, 4 ply minimum yarn

1.2.2 Type II. Type II mopheads shall be looped end mopheads of the following compositions and classes as specified (see 6.2).

1.2.2.1 Compositions.

Composition A - Cotton-blend yarn.
Composition B - Rayon or rayon-blend yarn.

1.2.2.2 Classes.

Class 1 - Cloth headband.
Class 2 - Synthetic mesh headband.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: General Services Administration, Federal Supply Service, General Products Commodity Center, General Engineering Group (7FXEE), 819 Taylor St., Fort Worth, TX 76102

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FSC 7920

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1.2.3 Sizes. Mopheads shall be of the following sizes as specified (see 6.2).

Size 9 - 9 pounds per dozen mopheads.
Size 12 - 12 pounds per dozen mopheads.
Size 15 - 15 pounds per dozen mopheads.
Size 16.5 - 16-1/2 pounds per dozen mopheads.
Size 18 - 18 pounds per dozen mopheads.
Size 20 - 20 pounds per dozen mopheads.
Size 24 - 24 pounds per dozen mopheads.

2. APPLICABLE DOCUMENTS

2.1 The documents referenced in this specification shall be the issues in effect on the date of the invitation for bids or request for proposal unless otherwise specified. These documents form a part of this commercial item description to the extent specified. In the event that there is a conflict between this commercial item description and a document referenced herein, this commercial item description shall take precedence.

Federal Standards:

FED-STD-376 - Preferred Metric Units For General use by the Federal Government
FED-STD-751 - Stitches, Seams, and Stitching

Copies of Federal Standards are available from the General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

Copies of Military Standards are available from Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

Industry Standards:

Technical Manual of the American Association of Textile Chemists and Colorists (AATCC):
AATCC Test Method 20 Fiber Analysis, Qualitative
AATCC Test Method 20A Fiber Analysis, Quantitative
AATCC Test Method 135 Dimensional Change in Automatic Home Laundering of Woven and Knitted Fabrics

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Copies of AATCC test methods are available from the American Association of Textile Chemists and Colorists, AATCC, 1 Davis Drive, P.O. Box 12215, Research Triangle Park, NC 27709, 919/549-8141.

ASTM D 204	Standard Methods of Testing Sewing Threads
ASTM D 2256	Standard Test Method for Tensile Properties of Yarns by the Single-Strand Method
ASTM D 3775	Standard Test Method of Fabric Count of Woven Fabric
ASTM D 3776	Standard Test Method for Mass Per Unit Area (Weight) of Woven Fabrics

Copies of ASTM standards are available from ASTM, 1916 Race Street, Philadelphia, PA 19103.

3. REQUIREMENTS

3.1. Materials.

3.1.1 Yarns.

3.1.1.1 Type I. The yarns for Type I mopheads shall be cotton or cotton blended with rayon, polyester, nylon, acrylic, or a combination of these fibers when tested in accordance with 4.5.1.1. The blended yarns shall be of an intimate fiber blend with a cotton content of not less than 30 percent by weight when tested in accordance with 4.5.1.2. The yarns shall be free of trash and foreign material, shall have cut ends, and shall meet the requirements as specified in table I when tested.

Table I. Physical Requirements, Type I Yarns

Characteristics	Style 1	Style 2	Style 3
Plies	8 (+/- 2)	4 minimum	4 minimum
Yards per pound	95 minimum	190 minimum	250 (+/- 25)
Turns per inch (tolerance)	1-1/4 (+/- 1/4)	1-1/2 (+/- 1/4)	2-1/2 (+/- 1/4)
Breaking strength, pounds	50 minimum	25 minimum	18 minimum

3.1.1.2 Type II.

3.1.1.2.1 Composition A yarns. The yarns for Type II, composition A mopheads shall be cotton blended with rayon, polyester, nylon, acrylic, or a combination of these fibers when tested in accordance with 4.5.1.1. The blended yarns shall be an intimate fiber blend with a cotton content not less than 30 percent or more than 60 percent by weight when tested in accordance with 4.5.1.2.

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3.1.1.2.2 Composition B yarns. The yarns for Type II, Composition B mopheads shall be 100 percent rayon or rayon blended with polyester, nylon, acrylic, or a combination of these fibers when tested in accordance with 4.5.1.1. The blended yarns shall contain, in intimate fiber blend, not less than 30 percent rayon by weight when tested in accordance with 4.5.1.2.

3.1.1.2.3 Physical requirements of type II mop yarns. The mop yarns shall be free of trash and foreign material, have continuous looped ends, and shall meet the requirements specified in table II when tested.

Table II. Physical Requirements, Type II Yarns

Plies	4 minimum
Yards per pound	150 maximum
Turns per inch	1-1/4 (+/- 1/4)
Breaking strength, pounds	50 minimum

3.1.1.2.4 Color. The color of the yarns for Type II mopheads shall be natural, orange, green, yellow, or blue, as specified (see 6.2). The shade shall be at the option of the manufacturer. The colors shall not fade when tested in accordance with 4.5.3.

3.1.1.3 Nonfibrous material. The yarns for all mopheads shall contain not more than 5 percent sizing, finishing or other nonfibrous materials when tested in accordance with 4.5.1.2.

3.1.2 Headbands.

3.1.2.1 Type I and Type II, Class 1. The cloth used for all the type I headbands and for the type II, class 1 headband shall have a fiber content of cotton, cotton-synthetic, or synthetic. The cloth shall weigh not less than 0.25 ounces per linear yard based on a 1 inch width, the weights for other widths shall be proportional (see 4.5.2). The finished headband may be woven tape, needle loom constructed tape, or cut strips of cloth. The edges of the band shall be woven, selvaged, or if cut tape shall be turned under and securely stitched with a 1/8 inch stitching gauge using nylon, polyester, or cotton thread as specified by 3.1.4. The headband on the end item mophead shall be 1-1/4 inch wide minimum.

3.1.2.2 Type II, Class 2. Type II, class 2 mopheads shall have a synthetic mesh headband. The synthetic mesh for the type II, class 2 mophead headbands shall be of synthetic fibers, scoured and dyed, woven or knitted into an open pattern, coated with a thermoplastic resin, and cured at 375 degrees Fahrenheit. The finished mesh shall weigh 5.4 ounces per square yard (plus or minus 0.5 ounce). The finished mesh shall be colorfast when tested in accordance with 4.5.3. The mesh headband shall be 3-3/4 inches in width minimum.

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3.1.3 Tailbands.

3.1.3.1 Type I, Style 2 and Type II, Class 1. The woven tape for the tailband for type I style 2 and type II, class 1 mopheads shall be of one piece, 1/2 inch to 3/4 inch wide, with selvaged or woven edges (needle loom construction is acceptable). The tape shall be of cotton, cotton-synthetic, or synthetic, bleached or unbleached, twill weave, texture based on not less than 90 or more than 96 warp yarns per inch and not less than 36 filling yarns per inch; except for needle loom constructed tape which shall have not less than 48 single ply filling yarns laid double per inch. If 2 ply warp yarns are used there shall be not less than 60 or more than 65 yarns per inch.

3.1.3.2 Type II, Class 2. The tailband for the type II, class 2 mophead shall be as specified in 3.1.3.1 or shall match the mesh used in the headband at the manufacturer's option.

3.1.4 Thread.

3.1.4.1 Cotton. The cotton sewing thread, when used for bar tacking the ends of the tapes, shall have a natural or white finish, or shall match the color of the mop at the manufacturers option. The cotton sewing thread shall have a weight of not less than 8000 yards per pound or more than 10,000 yards per pound, and shall have a breaking strength of 2.1 pounds minimum when tested in accordance with Table VI.

3.1.4.2 Nylon and polyester. The nylon or polyester thread shall be of a natural color or shall match the color of the mop at the manufacturers option. The nylon or polyester thread shall be bonded, multicord content, and have a soft finish. The nylon or polyester thread shall have a weight of not less than 2600 yards per pound or more than 3600 yards per pound, and shall have a breaking strength of 17 pounds minimum.

3.2 Construction.

3.2.1 Headband. All mopheads shall have a one-piece headband of the specified material completely encircling the middle of the evenly distributed yarns, with an overlap of 2 inches minimum. The centerline of the headband shall be within plus or minus 1/2 inch of the midpoint of the yarns. The yarns and the headband shall be securely sewn together with nylon or polyester thread as specified in 3.1.4.2 using stitch types 301 or 401 of FED-STD-751. The finished mophead shall be 6 inches wide, plus or minus 3/8 inch, at the headband prior to laundering. The number of rows of stitching shall be as specified. The outer rows of stitching shall be 1/16 inch to 1/2 inch from the edge of the band. The rows of stitching shall come no closer than 1/16 inch to each other and shall not cross.

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3.2.1.1 Type I headband. The type I headband shall be secured with a minimum of 4 rows of stitching, 6 stitches per inch minimum.

3.2.1.2 Type II headband. The type II, class 1 headband shall be secured with a minimum of 2 rows of stitching, 6 stitches per inch minimum. The type II, class 2 mophead headband shall be secured with a minimum of 8 rows of stitching, 6 stitches per inch minimum.

3.2.2 Tailband. The type I, style 2, and all type II mopheads shall have a tailband of the specified material superimposed on one side of the mop yarns and securely stitched to the yarns with not less than two equally spaced rows of stitching. The tailband shall maintain the yarns in a side-by-side relationship in a flat layer. The tailband shall be located at a distance of 1-1/2 to 2 inches from the end of the yarns, as measured from the edge of the tailband closest to the end of the yarns. The rows of stitching shall not cross or come within 1/16 inch of each other or the edge of the band. The sewing thread shall be nylon or polyester as specified in 3.1.4.2. The stitch shall be stitch type 301 or 401 of FED-STD-751, 6 stitches per inch minimum. Both ends of the tailband shall be pulled around the end yarns a minimum of 1 inch and bar tacked through the tailband with not less than 28 stitches per bar tack using cotton, nylon, or polyester thread as specified in 3.1.4.

3.2.3 Physical requirements of finished mopheads. The finished Type I and Type II mopheads shall conform to the requirements of table III and table IV respectively prior to laundering.

Table III. Physical Requirements of Type I Mopheads.

Size	9	12	15	18	20	24
Pounds/dozen, minimum	9	12	15	18	20	24
Yarns per mophead, minimum						
Style 1	90	100	120	130	140	165
Style 2	150	200	225	250	275	300
Style 3	---	----	240	---	---	335
Length of yarns, inches, minimum						
Style 1	26	31.5	33	37	38	40
Style 2	26	31	35	37	39	42
Style 3	--	--	35	--	--	40

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Table IV. Physical Requirements of the Type II Mopheads.

Size	12	15	16.5	18
Pounds per dozen, minimum	12	15	16-1/2	18
Length, inches, +/- 3 inches	34	39	41	43
Width at tailband, inches, +/- 2 inches	20	22	24	26

3.2.4 Shrinkage. The finished type II mopheads shall not shrink more than 20 percent when tested as specified in 4.5.3.

3.3 Special Marking. The headband of each mophead shall be indelibly marked with the National Stock Number (NSN) in letters not less than 1/8 inch in height.

3.4 Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this commercial item description unless otherwise specified.

3.5 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified in this specification. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections or tests set forth in the specifications where such action is deemed necessary to assure that supplies and services conform to the prescribed requirements. The contractor is responsible for insuring that components and materials used were manufactured, examined, and tested in accordance with referenced specifications and standards.

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4.2 Certificates of conformance. Unless otherwise specified, certificates of conformance may be acceptable in lieu of testing of end items. These certificates shall be based on the testing of the component materials and may be performed by the component material supplier.

4.3 Sampling. Sampling shall be in accordance with MIL-STD-105, the inspection level and the acceptable quality level (AQL) shall be as specified.

4.4 Visual and dimensional examination. Visual and dimensional examination of the end item shall be done on a lot by lot basis. The sample unit shall be one mophead, the lot shall be all mopheads of the same type, style, composition and class submitted for inspection at one time. The inspection level shall be I, the AQL shall be 4.0 percent defective.

Table V. Visual and Dimensional Defects.

Examine:	Defect:
Mop yarn	Not free of trash.
	Less than required ply.
	Type II - Not looped ends (one piece not looped will fail sample).
	Type II - Not specified color.
	Less than required length.
	Not evenly distributed across width of headband.
	Not in flat layer and evenly distributed by tailband, when applicable.
Headband	More than one piece.
	Band not minimum width.
	Less than 2 inches overlap.
	Not specified material.
	Not minimum number of rows of stitching.
	Incorrect stitch type.
	Band not equidistant from the yarn ends.
	Sewn by less than six stitches per inch.
	Outer rows of stitching not required distance from edges of band.
	Rows of sewing are less than 1/16 inch apart at any point.
Tailband (when required)	Less than two rows of stitching.
	Less than six stitches per inch.
	Stitching runs off band.
	Incorrect stitch type.
	Ends of band not turned around end yarns and bar tacked as specified.
	More than one piece of tape.
	Not required width.
	Edge of tape not required distance from ends of yarns.
	Not retaining all yarns (one loose yarn fails sample.

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4.4.1 Examination of the end item for defects in weight. This shall be determined on a lot by lot basis. The sample unit for this examination shall be 12 mopheads taken at random throughout the lot. The inspection level shall be S-2, AQL 6.5 percent defective. The 12 samples shall be weighed for conformance with the applicable requirements.

4.5 Sampling for tests. Sampling for tests shall be inspection level S-2, AQL 6.5 percent defective.

4.5.1 Yarn tests.

4.5.1.1 Fiber identification. The yarn composition shall be tested in accordance with the American Association of Textile Chemists and Colorists (AATCC) Test Method 20. The content of the yarns shall be as specified in 3.1.1.1 or 3.1.1.2 as applicable.

4.5.1.2 Yarn percentages. The samples shall be tested in accordance with AATCC Test Method 20A.

4.5.1.3 Yarn breaking strength. Each sample unit shall have 4 yarns tested for conformance to the breaking strength requirements specified in tables I or II as applicable. Each sample yarn shall be tested in accordance with ASTM D 2256, Option A, Condition 1. Failure of any of the tested yarns to meet the requirements shall fail the lot.

4.5.2 Headband and tailband tests. The material of the headband and tailband shall be tested in accordance with table VI. Component testing is acceptable.

Table VI. Headband & tailband tests.

Component - Test For	Requirement Para.	Test method
Headband cloth		
Thread count	3.1.2.1	ASTM D 3775
Weight	3.1.2.1	ASTM D 3776
Tailband cloth		
Thread count	3.1.3	ASTM D 3775
Mesh Headband		
Weight	3.1.2.2	ASTM D 3776
Thread		
Strength	3.1.4.1 & 3.1.4.2	ASTM D 204, Single Strand, Dry Breaking Strength

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4.5.3 End item shrinkage and colorfastness. This test shall be performed on a lot-by-lot basis. The type II samples shall be tested for shrinkage in accordance with AATCC test Method 135, except that the samples shall go through the wash and drying cycles 5 times total. The machine cycle shall be cotton sturdy, wash temperature V. The tumble dry setting shall be Cotton/Sturdy. The samples shall not show a dimensional change over 20 percent (see 3.2.4). The washed samples shall be compared with an unwashed sample from the same lot, the washed samples shall not show significant color change (see 3.1.1.2.4).

4.6 Examination for preparation for delivery. A random sample of shipping containers shall be selected from each lot for examination of the unit, intermediate, and shipping containers (as applicable) for conformance with the packaging, packing, and marking required in the contract or order. Samples shall be selected in accordance with MIL-STD-105, inspection level S-2, acceptable quality level (AQL) 6.5 defects per hundred units.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The packaging, packing, and marking shall be as specified in the contract or purchase order.

5.2 Headband marking. The headband of each mophead shall be marked as specified in paragraph 3.3.

6. NOTES

6.1 Intended use. The mopheads described by this specification are intended for wet-cleaning of floors using water and strong cleaning solutions. The type II mopheads are designed to allow multiple high temperature machine washings without excessive shrinkage, fraying, or tangling.

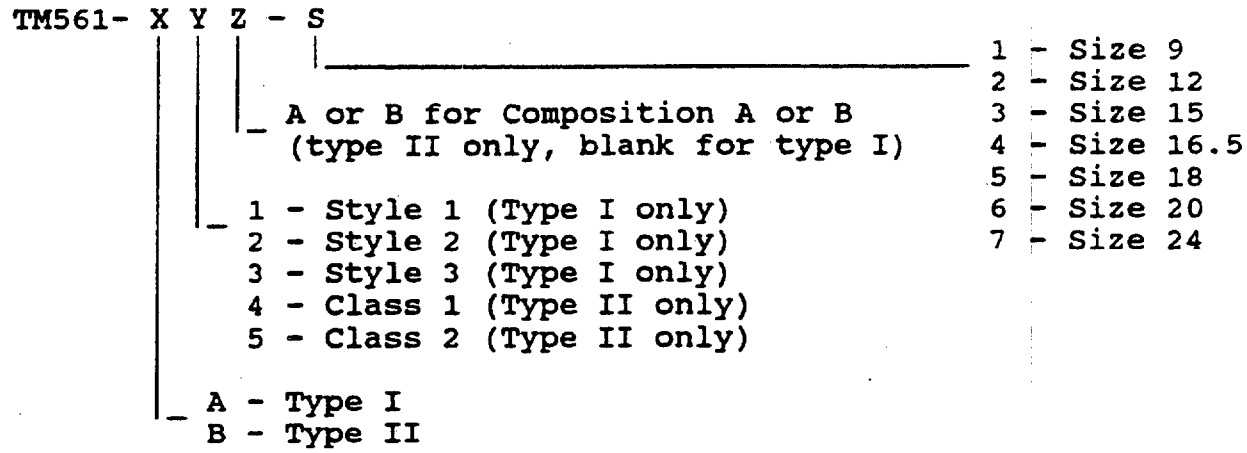
6.2 Ordering data. Purchasers should select the preferred options permitted here and include this information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, and size (see 1.2).
(c) Style (for type I only, see 1.2).
(d) Composition, class, and color (see 3.1.1.2.4) (for type II only, see 1.2).
(e) Packaging, packing, and markings required.

6.3 Supersession data. Type I contains the mopheads available under the "F" revision to this specification. All National Stock Numbers (NSN's) for wide band mopheads, specified as Type II of T-M-561F, have been cancelled. Mopheads designated Type II in this specification are now of looped end design, which at the time of the printing of this document had no NSN's assigned.

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6.4 Part numbering. Purchase descriptions using this document shall use the following part numbering system:



6.5 National stock numbers. The following NSN's have been identified as using this document (as of the printing of this document no NSN's exist for Type II mopheads):

NSN	Part Number (see 6.4)
7920-00-141-5544	TM561-A1-7
7920-00-141-5547	TM561-A1-5
7920-00-141-5548	TM561-A1-6
7920-00-141-5549	TM561-A1-1
7920-00-141-5550	TM561-A1-3
7920-00-171-1148	TM561-A1-2
7920-00-205-0425	TM561-A3-3
7920-00-205-0426	TM561-A3-7
7920-00-245-8290	TM561-A2-1
7920-00-926-5493	TM561-A2-2
7920-00-926-5494	TM561-A2-3
7920-00-926-5495	TM561-A2-5
7920-00-926-5496	TM561-A2-6
7920-00-926-5497	TM561-A2-7

MILITARY INTERESTS

NONE: DoD has determined that no military activity has an official interest in this Federal Specification.

Preparing Activity
GSA-FSS

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